

Assignment 1: Developing a Machine Learning App

Branden Labrador

AI Course Assignment

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Name and Purpose of the Application

The application is called Gold Price Predictor. Its purpose is to use historical gold price data to estimate a future monthly gold price. The program is written in Python and uses a supervised machine learning model because the dataset contains known historical prices that the model can learn from. The project focuses on clear data loading, simple preprocessing, model training, results, and a basic plot.

Algorithms Used

The application uses a Random Forest Regressor. This algorithm was selected because it can handle regression problems, which means it can predict a numeric value such as a commodity price. Random Forest also works well for beginner machine learning projects because it does not require advanced mathematical setup from the user. The model learns patterns from the year, month, and time order of the historical data to estimate the gold price.

Dataset Information

The dataset is a supplied CSV file named `gold_data.csv`. The file follows the DataHub Gold Prices dataset format, which uses a Date column and a Price column for monthly gold prices in U.S. dollars per troy ounce. The DataHub Gold Prices page describes the dataset as monthly gold prices in USD since 1833, sourced from historical gold records and World Bank commodity market data. For this project, the application reads the CSV directly from the project folder instead of downloading data automatically.

Libraries, Toolkits, and Frameworks

The project uses Pandas to read and prepare the CSV file, Matplotlib to create the graph, and Scikit-learn to train and evaluate the machine learning model. Pandas is used for data cleaning and feature creation. Matplotlib is used to visualize actual and predicted gold prices. Scikit-learn provides the Random Forest Regressor, the train-test split function, and the model evaluation metrics.

Application Design and Implementation

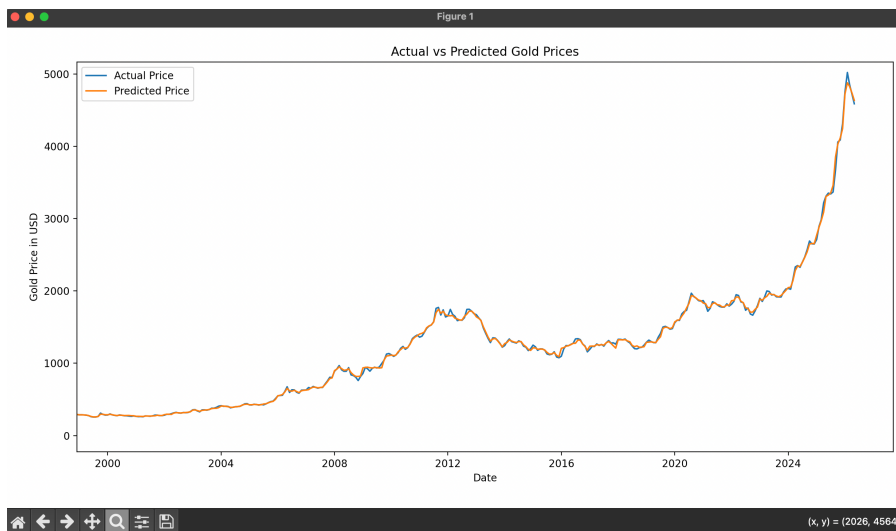
The application starts by loading `gold_data.csv`. It then removes missing values and converts the Date column into a usable date format. From that date, the program creates three simple features: Year, Month, and Time_Index. Year and Month help the model understand the time period, while Time_Index gives the model the order of the monthly records from oldest to newest. The program then separates the data into training and testing sets, trains the Random Forest model, evaluates the results, predicts the next month, and creates a chart comparing actual and predicted prices.

Instructions for Running the App

To run the application, keep `gold_predictor.py`, `gold_data.csv`, and `requirements.txt` in the same folder. First, install the required libraries by running `pip install -r requirements.txt`. Then run `python3 gold_predictor.py` from the project folder. The program will print the first rows of the dataset, show model results, print a future price prediction, and save a chart named `gold_prediction_chart.png`.

Results

When the application was tested, it loaded the supplied CSV file, trained the Random Forest model, and produced model evaluation results. The model reported a Mean Absolute Error of 5.53 and an R-squared score of 1.00 on the sample run. The predicted price for the next month in the dataset was \$4,611.95. These values show that the model followed the historical pattern in the dataset closely, although this does not mean the model can perfectly predict real market prices.



```
[brandens-MacBook-Air:gold_price_predictor_datahub_entry brandenlabrador$ python3 gold_predictor.py
First five rows of the dataset:
  Date      Price
0  1833-01-01  18.93
1  1833-02-01  18.93
2  1833-03-01  18.93
3  1833-04-01  18.93
4  1833-05-01  18.93

Model Results
Mean Absolute Error: 5.53
R-squared Score: 1.0

Future Gold Price Prediction
Prediction Month: 2026-06
Predicted Price: $4611.95
```

Discussion and Insights

The model performed well on the dataset because the input features are based on time and the gold price series follows a clear overall trend. However, this project has important limitations. Gold prices are affected by inflation, interest rates, currency movements, investor behavior, global events, and supply and demand. This entry-level model only uses date-based features, so it does not understand those outside economic factors. A more advanced version could include additional macroeconomic variables, compare several algorithms, or use more detailed daily price data.

References

DataHub. (2026). Gold prices. <https://datahub.io/core/gold-prices>

Scikit-learn developers. (2026). RandomForestRegressor. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>

The pandas development team. (2026). pandas documentation. <https://pandas.pydata.org/docs/>

Hunter, J. D., Dale, D., Firing, E., Droettboom, M., and Matplotlib Development Team. (2026). Matplotlib documentation. <https://matplotlib.org/stable/>